Recursive Problems and Dynamic Programming

A General Intertemporal Problem

Consider the following problem:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{T}} \quad U\left(k_0, k_1, ..., k_{T+1}, c_0, c_1, ..., c_T\right)$$

s.t.

$$F_0\left(k_0, k_1, ..., k_{T+1}, c_0, c_1, ..., c_T\right) = 0$$

$$F_1\left(k_0, k_1, ..., k_{T+1}, c_0, c_1, ..., c_T\right) = 0$$

$$\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots$$

$$F_T\left(k_0, k_1, ..., k_{T+1}, c_0, c_1, ..., c_T\right) = 0$$

$$k_0 \;\; given$$

where $c_t \in \mathbb{R}$, $k_t \in \mathbb{R}$ (disregard non-negativity constraints, just for simplicity).

The solution of the problem will be of the form:

$$c_0 = H_0(k_0)$$
$$\dots\dots\dots\dots\dots\dots$$
$$c_T = H_T(k_0)$$

$$k_1 = J_1(k_0)$$
$$\dots\dots\dots\dots\dots\dots$$
$$k_{T+1} = J_{T+1}(k_0)$$

Assume the problem is "nicely behaved" and form the Lagrangian:

$$\mathcal{L} = U\big(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T\big)$$

$$+\lambda_0 F_0\big(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T\big)$$

$$+\lambda_1 F_1\big(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T\big)$$

$$+ \dots$$

$$+ \lambda_T F_T\big(k_0, k_1, \dots, k_{T+1}, c_0, c_1, \dots, c_T\big)$$

First-Order Conditions (FOC)

Notation: $k \equiv (k_0, k_1, ..., k_{T+1})$, $c \equiv (c_0, c_1, ..., c_T)$

$$
\begin{array}{ll}
\frac{\partial \mathcal{L}}{\partial c_0} = 0 \Rightarrow & \frac{\partial U(k,c)}{\partial c_0} + \sum_{s=0}^{T} \lambda_s \frac{\partial F_s(k,c)}{\partial c_0} = 0 \\
\ldots\ldots\ldots\ldots & \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\
\frac{\partial \mathcal{L}}{\partial c_T} = 0 \Rightarrow & \frac{\partial U(k,c)}{\partial c_T} + \sum_{s=0}^{T} \lambda_s \frac{\partial F_s(k,c)}{\partial c_T} = 0 \\
 & \\
\frac{\partial \mathcal{L}}{\partial k_1} = 0 \Rightarrow & \frac{\partial U(k,c)}{\partial k_1} + \sum_{s=0}^{T} \lambda_s \frac{\partial F_s(k,c)}{\partial k_1} = 0 \\
\ldots\ldots\ldots\ldots & \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\
\frac{\partial \mathcal{L}}{\partial k_{T+1}} = 0 \Rightarrow & \frac{\partial U(k,c)}{\partial k_{T+1}} + \sum_{s=0}^{T} \lambda_s \frac{\partial F_s(k,c)}{\partial k_{T+1}} = 0 \\
 & \\
\frac{\partial \mathcal{L}}{\partial \lambda_0} = 0 \Rightarrow & F_0(k, c) = 0 \\
\ldots\ldots\ldots\ldots & \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\
\frac{\partial \mathcal{L}}{\partial \lambda_T} = 0 \Rightarrow & F_T(k, c) = 0
\end{array}
$$

We have a system of $3(T+1)$ equations to solve for $3(T+1)$ unknowns, $c_0, ..., c_T, k_1, ..., k_{T+1}, \lambda_0, ..., \lambda_T$, in terms of k_0.

Notice that, in general, the system has to be solved *simultaneously* for the $3(T+1)$ unknowns.

Special Case: A Recursive Problem

Assume:

$$U(k,c) = u_0(k_0, c_0) + ... + u_T(k_T, c_T) + S(k_{T+1})$$

$$F_t(k,c) = G_t(k_t, c_t) - k_{t+1}, \quad t \in \{0, ..., T\}$$

With these assumptions, our problem becomes:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{T}} \quad u_0(k_0, c_0) + ... + u_T(k_T, c_T) + S(k_{T+1})$$

s.t.

$$k_1 = G_0(k_0, c_0)$$

$$k_2 = G_1(k_1, c_1)$$

....

$$k_{T+1} = G_T(k_T, c_T)$$

$$k_0 \;\; given$$

c_0, $c_1, \ldots,$ c_T : control variables

k_0, $k_1, \ldots,$ k_{T+1} : state variables

$u_t(k_t, c_t)$: one-period return function at t

$S(k_{T+1})$: scrap value

$k_{t+1} = G_t(k_t, c_t)$: transition equation at t

The special time-separable structure of this problem implies that, for given k_τ, c_τ influences the future values of the state variable $\{k_{t+1}\}_{t=\tau}^{T}$ and the current and future returns $\{u_t(k_t, c_t)\}_{t=\tau}^{T}$ but it does not influence their past values, $\{k_{t+1}\}_{t=0}^{\tau-1}$ and $\{u_t(k_t, c_t)\}_{t=0}^{\tau-1}$. We have:

$$
\begin{aligned}
k_{\tau+1} &= G_\tau(k_\tau, c_\tau) \\
k_{\tau+2} &= G_{\tau+1}(k_{\tau+1}, c_{\tau+1}) \\
&= G_{\tau+1}(G_\tau(k_\tau, c_\tau), c_{\tau+1}) \\
k_{\tau+3} &= G_{\tau+2}(k_{\tau+2}, c_{\tau+2}) \\
&= G_{\tau+2}(G_{\tau+1}(G_\tau(k_\tau, c_\tau), c_{\tau+1}), c_{\tau+2}) \\
&\ldots.
\end{aligned}
$$

and

$$
\begin{aligned}
&u_\tau(k_\tau, c_\tau) \\
&u_{\tau+1}(k_{\tau+1}, c_{\tau+1}) = u_{\tau+1}(G_\tau(k_\tau, c_\tau), c_{\tau+1}) \\
&u_{\tau+2}(k_{\tau+2}, c_{\tau+2}) = u_{\tau+2}(G_{\tau+1}(G_\tau(k_\tau, c_\tau), c_{\tau+1}), c_{\tau+2}) \\
&\ldots.
\end{aligned}
$$

Therefore, k_τ constitutes a complete description of the current position of the system (past values $\{c_t, k_t\}_{t=0}^{\tau-1}$ add no information beyond that contained in k_τ; this is why we call k_τ the state variable at time τ). We show below that this key property gives the problem a recursive structure.

The Lagrangian is:

$$\begin{aligned}\mathcal{L} = & u_0(k_0,c_0)+u_1(k_1,c_1)+...+u_T(k_T,c_T)+S(k_{T+1}) \\ & +\lambda_0[G_0(k_0,c_0)-k_1]+\lambda_1[G_1(k_1,c_1)-k_2]+ .... \\ & +\lambda_{T-1}\left[G_{T-1}(k_{T-1},c_{T-1})-k_T\right]+\lambda_T\left[G_T(k_T,c_T)-k_{T+1}\right]\end{aligned}$$

First-Order Conditions (FOC)

$$\frac{\partial \mathcal{L}}{\partial c_0} = 0 \Rightarrow \qquad \frac{\partial u_0(k_0,c_0)}{\partial c_0} + \lambda_0 \frac{\partial G_0(k_0,c_0)}{\partial c_0} = 0$$

....

$$\frac{\partial \mathcal{L}}{\partial c_T} = 0 \Rightarrow \qquad \frac{\partial u_T(k_T,c_T)}{\partial c_T} + \lambda_T \frac{\partial G_T(k_T,c_T)}{\partial c_T} = 0$$

$$\frac{\partial \mathcal{L}}{\partial k_1} = 0 \Rightarrow \qquad \frac{\partial u_1(k_1,c_1)}{\partial k_1} - \lambda_0 + \lambda_1 \frac{\partial G_1(k_1,c_1)}{\partial k_1} = 0$$

....

$$\frac{\partial \mathcal{L}}{\partial k_T} = 0 \Rightarrow \qquad \frac{\partial u_T(k_T,c_T)}{\partial k_T} - \lambda_{T-1} + \lambda_T \frac{\partial G_T(k_T,c_T)}{\partial k_T} = 0$$

$$\frac{\partial \mathcal{L}}{\partial k_{T+1}} = 0 \Rightarrow \quad S'(k_{T+1}) - \lambda_T = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_0} = 0 \Rightarrow \qquad k_1 = G_0(k_0, c_0)$$

....

$$\frac{\partial \mathcal{L}}{\partial \lambda_T} = 0 \Rightarrow \qquad k_{T+1} = G_T(k_T, c_T)$$

Now we'll show that the system of first-order conditions can be solved recursively. A *recursive solution* takes the form:

$$
\begin{aligned}
c_t &= h_t(k_t) \\
k_{t+1} &= g_t(k_t) \equiv G_t(k_t, h_t(k_t))
\end{aligned}
$$

where the functions $h_t(\cdot)$ are called *policy functions* (also called, *optimal feedback rules*).

Iterations over the functions above, starting from the initial value of k_0 at $t = 0$, give the solution we are looking for:

$$c_0 = H_0(k_0) \equiv h_0(k_0)$$

$$k_1 = J_1(k_0) \equiv G_0(k_0, H_0(k_0))$$

$$c_1 = H_1(k_0) \equiv h_1(J_1(k_0))$$

$$k_2 = J_2(k_0) \equiv G_1(J_1(k_0), H_1(k_0))$$

$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

$$c_T = H_T(k_0) \equiv h_T(J_T(k_0))$$

$$k_{T+1} = J_{T+1}(k_0) \equiv G_T(J_T(k_0), H_T(k_0))$$

Start at $t = T$.

The FOC for period T are:

$$\frac{\partial u_T(k_T, c_T)}{\partial c_T} + \lambda_T \frac{\partial G_T(k_T, c_T)}{\partial c_T} = 0$$

$$\lambda_T = S'(k_{T+1})$$

$$k_{T+1} = G_T(k_T, c_T)$$

Notice that we have a system of three equations that can be solved for the three unknowns, c_T, k_{T+1}, and λ_T, in terms of k_T. We get:

$$c_T = h_T(k_T)$$

$$k_{T+1} = g_T(k_T) \equiv G_T(k_T, h_T(k_T))$$

$$\lambda_T = \ell_T(k_T)$$

Now go one step back to $t = T - 1$.

The FOC for period $T - 1$ are:

$$\frac{\partial u_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} + \lambda_{T-1} \frac{\partial G_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} = 0$$

$$\frac{\partial u_T(k_T, c_T)}{\partial k_T} - \lambda_{T-1} + \lambda_T \frac{\partial G_T(k_T, c_T)}{\partial k_T} = 0$$

$$k_T = G_{T-1}(k_{T-1}, c_{T-1})$$

We know from our period-T FOC that $c_T = h_T(k_T)$ and $\lambda_T = \ell_T(k_T)$. Hence, we can write:

$$\frac{\partial u_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} + \lambda_{T-1} \frac{\partial G_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} = 0$$

$$\frac{\partial u_T(k_T, h_T(k_T))}{\partial k_T} - \lambda_{T-1} + \ell_T(k_T) \frac{\partial G_T(k_T, h_T(k_T))}{\partial k_T} = 0$$

$$k_T = G_{T-1}(k_{T-1}, c_{T-1})$$

This is a system of three equations that can be solved for the three unknowns, c_{T-1}, k_T, and λ_{T-1}, in terms of k_{T-1}. We get:

$$c_{T-1} = h_{T-1}(k_{T-1})$$

$$k_T = g_{T-1}(k_{T-1}) \equiv G_{T-1}(k_{T-1}, h_{T-1}(k_{T-1}))$$

$$\lambda_{T-1} = \ell_{T-1}(k_{T-1})$$

Now go one step back to $t = T - 2$.

The FOC for period $T - 2$ are:

$$\frac{\partial u_{T-2}(k_{T-2}, c_{T-2})}{\partial c_{T-2}} + \lambda_{T-2} \frac{\partial G_{T-2}(k_{T-2}, c_{T-2})}{\partial c_{T-2}} = 0$$

$$\frac{\partial u_{T-1}(k_{T-1}, c_{T-1})}{\partial k_{T-1}} - \lambda_{T-2} + \lambda_{T-1} \frac{\partial G_{T-1}(k_{T-1}, c_{T-1})}{\partial k_{T-1}} = 0$$

$$k_{T-1} = G_{T-2}(k_{T-2}, c_{T-2})$$

We know from the FOC for period $T-1$ that $c_{T-1} = h_{T-1}(k_{T-1})$ and $\lambda_{T-1} = \ell_{T-1}(k_{T-1})$. Hence, we can write:

$$\frac{\partial u_{T-2}(k_{T-2}, c_{T-2})}{\partial c_{T-2}} + \lambda_{T-2} \frac{\partial G_{T-2}(k_{T-2}, c_{T-2})}{\partial c_{T-2}} = 0$$

$$\frac{\partial u_{T-1}(k_{T-1}, h_{T-1}(k_{T-1}))}{\partial k_{T-1}} - \lambda_{T-2}$$

$$+\ell_{T-1}(k_{T-1}) \frac{\partial G_{T-1}(k_{T-1}, h_{T-1}(k_{T-1}))}{\partial k_{T-1}} = 0$$

$$k_{T-1} = G_{T-2}(k_{T-2}, c_{T-2})$$

This is a system of three equations that can be solved for the three unknowns, c_{T-2}, k_{T-1}, and λ_{T-2}, in terms of k_{T-2}. We get:

$$c_{T-2} = h_{T-2}(k_{T-2})$$

$$k_{T-1} = g_{T-2}(k_{T-2}) \equiv G_{T-2}(k_{T-2}, h_{T-2}(k_{T-2}))$$

$$\lambda_{T-2} = \ell_{T-2}(k_{T-2})$$

Keep going until $t = 0$

The FOC for period 0 are:

$$\frac{\partial u_0(k_0,c_0)}{\partial c_0} + \lambda_0 \frac{\partial G_0(k_0,c_0)}{\partial c_0} = 0$$

$$\frac{\partial u_1(k_1,c_1)}{\partial k_1} - \lambda_0 + \lambda_1 \frac{\partial G_1(k_1,c_1)}{\partial k_1} = 0$$

$$k_1 = G_0(k_0, c_0)$$

We know from the FOC for period 1 that $c_1 = h_1(k_1)$ and $\lambda_1 = \ell_1(k_1)$. Hence, we can write:

$$\frac{\partial u_0(k_0,c_0)}{\partial c_0} + \lambda_0 \frac{\partial G_0(k_0,c_0)}{\partial c_0} = 0$$

$$\frac{\partial u_1(k_1,h_1(k_1))}{\partial k_1} - \lambda_0 + \ell_1(k_1) \frac{\partial G_1(k_1,h_1(k_1))}{\partial k_1} = 0$$

$$k_1 = G_0(k_0, c_0)$$

This is a system of three equations that can be solved for the three unknowns, c_0, k_1, and λ_0, in terms of k_0. We get:

$$c_0 = h_0(k_0)$$

$$k_1 = g_0(k_0) \equiv G_0(k_0, h_0(k_0))$$

$$\lambda_0 = \ell_0(k_0)$$

and we are done.

Bellman Equations (BE)

Define the following sequence of *one-period* optimization problems, for $t \in \{0, 1, ..., T\}$:

$$W_t(k_t) \equiv \max_{\substack{c_t,\, k_{t+1} \\ \text{s.t.}}} \; u_t(k_t, c_t) + W_{t+1}(k_{t+1})$$

$$k_{t+1} = G_t(k_t, c_t)$$
$$k_t \;\; given$$

with

$$W_{T+1}(k_{T+1}) = S(k_{T+1})$$

We'll show that the solution to this sequence of one-period problems coincides with the solution to our original problem.

Consider period $t = T$.

Then:

$$W_T(k_T) \equiv \max_{c_T,\, k_{T+1}} \; u_T(k_T, c_T) + S(k_{T+1})$$

s.t.

$$k_{T+1} = G_T(k_T, c_T)$$
$$k_T \;\; given$$

The Lagrangian for this problem is:

$$\mathcal{L} = u_T(k_T, c_T) + S(k_{T+1}) + \lambda_T \left[G_T(k_T, c_T) - k_{T+1} \right]$$

FOC

$$\frac{\partial \mathcal{L}}{\partial c_T} = 0 \Rightarrow \quad \frac{\partial u_T(k_T, c_T)}{\partial c_T} + \lambda_T \frac{\partial G_T(k_T, c_T)}{\partial c_T} = 0$$

$$\frac{\partial \mathcal{L}}{\partial k_{T+1}} = 0 \Rightarrow \quad \lambda_T = S'(k_{T+1})$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_T} = 0 \Rightarrow \quad k_{T+1} = G_T(k_T, c_T)$$

Notice that this system coincides exactly with the period-T FOC of our original problem. Hence, we get:

$$c_T = h_T(k_T)$$

$$k_{T+1} = g_T(k_T) \equiv G_T(k_T, h_T(k_T))$$

$$\lambda_T = \ell_T(k_T) \equiv S'(G_T(k_T, h_T(k_T)))$$

Substituting into the objective function we get:

$$\begin{aligned} W_T(k_T) &= u_T(k_T, h_T(k_T)) + S(g_T(k_T)) \\ &= u_T(k_T, h_T(k_T)) + S(G_T(k_T, h_T(k_T))) \end{aligned}$$

Now consider period $t = T-1$.

Then:

$$W_{T-1}(k_{T-1}) \equiv \max_{c_{T-1},\, k_T} \quad u_{T-1}(k_{T-1}, c_{T-1}) + W_T(k_T)$$

$$\text{s.t.}$$

$$k_T = G_{T-1}(k_{T-1}, c_{T-1})$$

$$k_{T-1} \quad given$$

The Lagrangian for this problem is:

$$\mathcal{L} = u_{T-1}(k_{T-1}, c_{T-1}) + W_T(k_T) + \lambda_{T-1}\left[G_{T-1}(k_{T-1}, c_{T-1}) - k_T\right]$$

FOC

$$\frac{\partial \mathcal{L}}{\partial c_{T-1}} = 0 \Rightarrow \quad \frac{\partial u_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} + \lambda_{T-1}\frac{\partial G_{T-1}(k_{T-1}, c_{T-1})}{\partial c_{T-1}} = 0$$

$$\frac{\partial \mathcal{L}}{\partial k_T} = 0 \Rightarrow \quad W_T'(k_T) - \lambda_{T-1} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_{T-1}} = 0 \Rightarrow \quad k_T = G_{T-1}(k_{T-1}, c_{T-1})$$

We know from our period-T problem that

$$W_T(k_T) = u_T(k_T, h_T(k_T)) + S(G_T(k_T, h_T(k_T)))$$

Then:

$$W_T'(k_T) =$$

$$= \frac{\partial u_T(k_T, h_T(k_T))}{\partial k_T} + \frac{\partial u_T(k_T, h_T(k_T))}{\partial c_T} h_T'(k_T)$$
$$+ S'(G_T(k_T, h_T(k_T))) \times \left[\frac{\partial G_T(k_T, h_T(k_T))}{\partial k_T} + \frac{\partial G_T(k_T, h_T(k_T))}{\partial c_T} h_T'(k_T) \right]$$

$$= \frac{\partial u_T(k_T, h_T(k_T))}{\partial k_T}$$
$$+ \left[\frac{\partial u_T(k_T, h_T(k_T))}{\partial c_T} + S'(G_T(k_T, h_T(k_T))) \frac{\partial G_T(k_T, h_T(k_T))}{\partial c_T} \right] h_T'(k_T)$$
$$+ S'(G_T(k_T, h_T(k_T))) \frac{\partial G_T(k_T, h_T(k_T))}{\partial k_T}$$

$$= \frac{\partial u_T(k_T, h_T(k_T))}{\partial k_T} + \left[\frac{\partial u_T(k_T, h_T(k_T))}{\partial c_T} + \ell_T(k_T) \frac{\partial G_T(k_T, h_T(k_T))}{\partial c_T} \right] h_T'(k_T)$$
$$+ \ell_T(k_T) \frac{\partial G_T(k_T, h_T(k_T))}{\partial k_T} \quad \text{the term in } [\bullet] \text{ equals 0 by FOC of period } T$$

$$= \frac{\partial u_T(k_T, h_T(k_T))}{\partial k_T} + \ell_T(k_T) \frac{\partial G_T(k_T, h_T(k_T))}{\partial k_T}$$

Then, the period-$T-1$ FOC become:

$$\frac{\partial u_{T-1}(k_{T-1},c_{T-1})}{\partial c_{T-1}} + \lambda_{T-1}\frac{\partial G_{T-1}(k_{T-1},c_{T-1})}{\partial c_{T-1}} = 0$$

$$\frac{\partial u_T(k_T,h_T(k_T))}{\partial k_T} + \ell_T(k_T)\frac{\partial G_T(k_T,h_T(k_T))}{\partial k_T} - \lambda_{T-1} = 0$$

$$k_T = G_{T-1}(k_{T-1}, c_{T-1})$$

But this system coincides exactly with the period-$T-1$ FOC of our original problem. Hence:

$$c_{T-1} = h_{T-1}(k_{T-1})$$

$$k_T = g_{T-1}(k_{T-1}) \equiv G_{T-1}(k_{T-1}, h_{T-1}(k_{T-1}))$$

$$\lambda_{T-1} = \ell_{T-1}(k_{T-1})$$

Substituting into the objective function we get:

$$\begin{aligned} W_{T-1}(k_{T-1}) &= u_{T-1}(k_{T-1},h_{T-1}(k_{T-1}))+W_T(g_{T-1}(k_{T-1})) \\ &= u_{T-1}(k_{T-1},h_{T-1}(k_{T-1}))+W_T(G_{T-1}(k_{T-1},h_{T-1}(k_{T-1}))) \end{aligned}$$

We can keep going backward in time and show that, in each period, the solution coincides with the solution to our original problem.

Bottom Line: we can solve our original problem by solving the sequence of one-period problems defined by the Bellman Equations.

Another perspective on the recursive nature of the problem can be obtained from the following observations.

First, notice that:

$$
\begin{aligned}
W_0(k_0) &= \max_{c_0,\, k_1} \{u_0(k_0,c_0)+W_1(k_1)\} \\
&= \max_{c_0,\, k_1} \{u_0(k_0,c_0)+\max_{c_1,\, k_2} \{u_1(k_1,c_1)+W_2(k_2)\}\} \\
&= \max_{c_0,\, k_1} \{u_0(k_0,c_0)+\max_{c_1,\, k_2} \{u_1(k_1,c_1) \\
&\quad +\max_{c_2,\, k_3} \{u_2(k_2,c_2)+W_3(k_3)\}\}\} \\
&= \max_{c_0,\, k_1} \{u_0(k_0,c_0)+\max_{c_1,\, k_2} \{u_1(k_1,c_1) \\
&\quad +\max_{c_2,\, k_3} \{u_2(k_2,c_2)+\max_{c_3,\, k_4} \{u_3(k_3,c_3)+W_4(k_4)\}\}\}\} \\
&= \max_{c_0,\, k_1} \{u_0(k_0,c_0)+\max_{c_1,\, k_2} \{u_1(k_1,c_1) \\
&\quad +\max_{c_2,\, k_3} \{u_2(k_2,c_2)+\max_{c_3,\, k_4} \{u_3(k_3,c_3) \\
&\quad +\ldots.+\max_{c_T,\, k_{T+1}} \{u_T(k_T,c_T)+S(k_{T+1})\}\ldots\}\}\}\}
\end{aligned}
$$

where, in each step, the maximization is understood to be subject to $k_{t+1} = G_t(k_t, c_t)$, with k_t given.

Second, we know the solutions to our two problems coincide. Then:

$$W_0(k_0) = \max_{\{c_t, k_{t+1}\}_{t=0}^{T}} \{u_0(k_0, c_0) + ... + u_T(k_T, c_T) + S(k_{T+1})\}$$

where the maximization is subject to $k_{t+1} = G_t(k_t, c_t)$ for all $t \in \{0, 1, ...\}$, with k_0 given.

Combining the two results:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{T}} \{u_0(k_0, c_0) + ... + u_T(k_T, c_T) + S(k_{T+1})\} =$$

$$\max_{c_0,\, k_1} \{u_0(k_0, c_0) + \max_{c_1,\, k_2} \{u_1(k_1, c_1) + \max_{c_2,\, k_3} \{u_2(k_2, c_2) +$$

$$\max_{c_3,\, k_4} \{u_3(k_3, c_3) + ... + \max_{c_T,\, k_{T+1}} \{u_T(k_T, c_T) + S(k_{T+1})\} ...\}\}\}\}$$

The expression above shows that, for our problem, it is legitimate to cascade the maximization operator. That is, the large optimization problem of the left-hand side can be broken up into $T+1$ smaller problems. First, the problem in the innermost brackets is solved and the optimal values of c_T and k_{T+1} are obtained for any value of k_T : $c_T = h_T(k_T)$ and $k_{T+1} = g_T(k_T)$. From these we obtain the optimized value $W_T(k_T)$. Then the problem in the second innermost brackets is solved, the optimizers being $c_{T-1} = h_{T-1}(k_{T-1})$ and $k_T = g_{T-1}(k_{T-1})$, with optimized value $W_{T-1}(k_{T-1})$. This process continues until the problem in the outermost brackets is reached.

The preceding argument implies the following principle, usually called the "Bellman's Principle of Optimality": An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

Bellman's Principle of Optimality (BPO)

Suppose the policy functions $\{c_t = h_t(k_t)\}_{t=0}^{T}$ solve

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{T}} \quad u_0(k_0, c_0) + ... + u_T(k_T, c_T) + S(k_{T+1})$$

$$\text{s.t.} \qquad k_{t+1} = G_t(k_t, c_t) \quad t \in \{0, 1, ..., T\}$$

$$k_0 \ \ given$$

Then $\{c_t = h_t(k_t)\}_{t=\tau}^{T}$ solve

$$\max_{\{c_t, k_{t+1}\}_{t=\tau}^{T}} \quad u_\tau(k_\tau, c_\tau) + ... + u_T(k_T, c_T) + S(k_{T+1})$$

$$\text{s.t.} \qquad k_{t+1} = G_t(k_t, c_t) \quad t \in \{\tau, \tau+1, ..., T\}$$

$$k_\tau \ given$$

for any $\tau \in \{0, 1, ..., T\}$.

Remark. BPO implies that, as time advances, there is no incentive to depart from the original plan. Hence, optimal policies have a *self-enforcing* character. Optimal policies that satisfiy BPO are said to be *time consistent*. This property is a consequence of the recursive character of the problem and will not characterize the solutions to more general problems.

Particular Case

$$u_t(k_t, c_t) = \beta^t u(k_t, c_t) \quad \text{with } \beta \in (0, 1)$$

Then, BE becomes:

$$W_t(k_t) \equiv \max_{c_t,\, k_{t+1}} \ \beta^t u(k_t, c_t) + W_{t+1}(k_{t+1})$$

$$\text{s.t.} \quad k_{t+1} = G_t(k_t, c_t)$$

$$k_t \ given$$

Define the *current* value function:

$$V_t(k_t) \equiv \frac{W_t(k_t)}{\beta^t}$$

Remark. For $t = 0$, $V_0(k_0) = W_0(k_0)$

Then:

$$
\begin{aligned}
V_t(k_t) &= \frac{W_t(k_t)}{\beta^t} \\
&= \frac{\max\limits_{c_t,\, k_{t+1}} \ \beta^t u(k_t,c_t)+W_{t+1}(k_{t+1})}{\beta^t} \\
&= \max_{c_t,\, k_{t+1}} \ \frac{\beta^t u(k_t,c_t)+W_{t+1}(k_{t+1})}{\beta^t} \\
&= \max_{c_t,\, k_{t+1}} \ u(k_t, c_t) + \frac{W_{t+1}(k_{t+1})}{\beta^t} \\
&= \max_{c_t,\, k_{t+1}} \ u(k_t, c_t) + \beta \frac{W_{t+1}(k_{t+1})}{\beta^{t+1}} \\
V_t(k_t) &= \max_{c_t,\, k_{t+1}} \ u(k_t, c_t) + \beta V_{t+1}(k_{t+1})
\end{aligned}
$$

Up to now, we've been dealing with the following problem:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{T}} \quad \sum_{t=0}^{T} u_t(k_t, c_t) + S(k_{T+1})$$

$$\text{s.t.} \qquad k_{t+1} = G_t(k_t, c_t) \quad t \in \{0, 1, ..., T\}$$

$$k_0 \ \ given$$

In general, the solution to this problem gives a different policy function $c_t = h_t(k_t)$ to be used at each date $t \in \{0, 1, ..., T\}$. This is a consequence of two features of the problem: the fact that the horizon is finite and the fact that the functions $u_t(\cdot)$ and $G_t(\cdot)$ have been permitted to depend on time in arbitrary ways.

We would like to discover contexts in which the same policy function $c_t = h(k_t)$ is used for each period t. These are called *time-invariant* or *stationary* policies. The previous remark suggests that we consider an *infinite-horizon* setup, and that we restrict the time dependence of $u_t(\cdot)$ and $G_t(\cdot)$ appropriately.

In addition to an infinite horizon, we'll assume:

$$
\begin{aligned}
u_t(k_t, c_t) &= \beta^t u(k_t, c_t) \quad \beta \in (0, 1) \\
G_t(k_t, c_t) &= G(k_t, c_t)
\end{aligned}
$$

An Infinite-Horizon Problem

Consider the problem:

$$\max_{\{c_t,k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(k_t,c_t) = u(k_0,c_0) + \beta u(k_1,c_1) + \beta^2 u(k_2,c_2)...$$

$$\text{s.t.} \qquad k_{t+1} = G(k_t, c_t) \quad t \in \{0, 1, ...\}$$

$$k_0 \;\; given$$

Notation: $\sum_{t=0}^{\infty} \beta^t u(k_t, c_t)$ denotes $\lim_{T\to\infty} \sum_{t=0}^{T} \beta^t u(k_t, c_t)$.

Define the period-1 continuation problem:

$$\max_{\{c_t,k_{t+1}\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} u(k_t,c_t) = u(k_1,c_1) + \beta u(k_2,c_2) + \beta^2 u(k_3,c_3)...$$

$$\text{s.t.} \qquad k_{t+1} = G(k_t, c_t) \quad t \in \{1, 2, ...\}$$

$$k_1 \;\; given$$

Notice that both problems have exactly the same form, except for the initial value of the state variable (the time invariance of the functions $u(\cdot)$ and $G(\cdot)$ is crucial for this). Moreover, this is true for arbitrary periods, t and $t+s$.

The property described above suggests that the solution to the original problem will be a time-invariant policy function

$$c_t = h(k_t)$$

and that the value function for the original problem and any of the continuation problems will be the same. Hence, for any $t \in \{0, 1, ...\}$, the Bellman Equation will be:

$$V(k_t) \equiv \max_{c_t,\ k_{t+1}} \ u(k_t, c_t) + \beta V(k_{t+1})$$

$$\text{s.t.} \quad k_{t+1} = G(k_t, c_t)$$

$$k_t \ \ given$$

Remark. To emphasize the irrelevance of the time subscript, some authors use c for the control variable, k for the current-period state and k' for its next-period value.

Substituting the constraint into the objective:

$$V(k_t) = \max_{c_t} \; u(k_t, c_t) + \beta V(G(k_t, c_t))$$

This is a *functional equation* (an equation in which the unknown is a function) to be solved for the function $V(\cdot)$. Notice that in the process of solving for $V(\cdot)$ we will be jointly solving for the policy function $h(\cdot)$.

There are different methods for solving the BE. We describe two of them.

1. *Value-function Iteration*

This method proceeds by constructing a sequence of value functions and associated policy functions. The sequence is created by iterating on

$$V_{j+1}(k) = \max_c \; u(k,c) + \beta V_j(G(k,c))$$

starting from some function V_0 (usually, $V_0 = 0$), and continuing until V_j has converged (*i.e.*, until $V_{j+1} = V_j$).

Remark. To show that this method works, one defines an operator T by $(TV)(k) = \max_c u(k,c) + \beta V(G(k,c))$. This operator maps V_j into $V_{j+1} : V_{j+1} = TV_j$. Under certain conditions, this operator is a contraction. Then, the contraction-mapping theorem ensures that T has a unique fixed point, $V = TV$. Moreover, the fixed point can be calculated by succesive iterations on $V_{j+1} = TV_j$ starting with some function V_0 (usually $V_0 = 0$).

2. *Guess and Verify*

This method involves guessing a solution V to the BE and verifying whether the guess is correct. The method of undetermined coefficients is used to verify the guess.

The Euler Equation

The previous solution methods require that we find the value function V. We show now that it may be possible to solve the problem without this knowledge.

Consider the Bellman Equation for the infinite-horizon problem:

$$V(k_t) = \max_{c_t} \ u(k_t, c_t) + \beta V(G(k_t, c_t))$$

If V were known (and differentiable), we could solve the problem in the right-hand side using standard calculus methods. The FOC for an interior solution is:

$$u_c(k_t, c_t) + \beta V'(G(k_t, c_t)) G_c(k_t, c_t) = 0$$

The equation above defines, implicitly, the policy function:

$$c_t = h(k_t)$$

Substituting into the objective we get:

$$V(k_t) = u(k_t, h(k_t)) + \beta V(G(k_t, h(k_t)))$$

Differentiating with respect to k_t we get:

$$\begin{aligned}
V'(k_t) &= \\
&= u_k(k_t,h(k_t)) + u_c(k_t,h(k_t))h'(k_t) \\
&\quad + \beta V'(G(k_t,h(k_t)))[G_k(k_t,h(k_t)) + G_c(k_t,h(k_t))h'(k_t)] \\
&= u_k(k_t,h(k_t)) + \beta V'(G(k_t,h(k_t)))G_k(k_t,h(k_t)) \\
&\quad + \underbrace{[u_c(k_t, h(k_t)) + \beta V'(G(k_t, h(k_t)))G_c(k_t, h(k_t))]}_{=0 \text{ by FOC}} h'(k_t)
\end{aligned}$$

Then:

$$V'(k_t) = u_k(k_t, h(k_t)) + \beta V'(G(k_t, h(k_t)))G_k(k_t, h(k_t))$$

This is the so-called Benveniste-Scheinkman (BS) formula, or Envelope Condition (since it can be obtained through a direct application of the Envelope Theorem).

In many problems, there is no unique way of defining states and controls, and several alternative definitions lead to the same solution of the problem. Suppose we manage to choose the state and control variables in such a way that the transition equation does not depend on k_t, so $k_{t+1} = G(c_t)$ [see the appendix for the more general case]. Then, $G_k = 0$, and the BS formula becomes:

$$\begin{aligned} V'(k_t) &= u_k(k_t, h(k_t)) \\ &= u_k(k_t, c_t) \end{aligned}$$

Shifting the expression above one period ahead we get:

$$V'(k_{t+1}) = u_k(k_{t+1}, c_{t+1})$$

From the FOC:

$$u_c(k_t, c_t) + \beta V'(k_{t+1}) G_c(k_t, c_t) = 0$$

Combining the two expressions above we obtain the *Euler Equation*:

$$u_c(k_t, c_t) + \beta u_k(k_{t+1}, c_{t+1}) G_c(k_t, c_t) = 0$$

From $k_{t+1} = G(c_t)$ we get (assuming G can be inverted): $c_t = G^{-1}(k_{t+1}) \equiv m(k_{t+1})$. Then, the Euler Equation becomes:

$$u_c(k_t, m(k_{t+1})) + \beta u_k(k_{t+1}, m(k_{t+2})) G_c(k_t, m(k_{t+1})) = 0$$

which is a second-order difference equation in k. This equation can be solved using two boundary conditions: the initial condition k_0, and an appropriate transversality condition (TVC).

Remark 1. Notice that we've managed to eliminate the value function V from our problem. The cost of doing this is that now we need to solve a nonlinear second-order difference equation. Moreover, in order to solve this equation we need to derive the appropriate TVC that will serve as our second boundary condition (more on this next class).

Remark 2. The Guess-Verify method can sometimes be used to solve the Euler Equation. Instead of guessing the form of the value function, we guess the form of the policy function $h(\bullet)$, and check that the Euler Equation is satisfied.

Remark 3. The solution to the Euler Equation will be an *open-loop* solution, since it will express k_t as a function of time only. Solutions of the form $c_t = h(k_t)$ and $k_{t+1} = g(k_t)$, like the ones we found earlier, are usually called *closed-loop* solutions.

Stochastic Dynamic Programming

Consider the (stationary) infinite-horizon stochastic problem:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \quad \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(k_t, c_t)$$

$$\text{s.t.} \qquad k_{t+1} = G(k_t, c_t, \varepsilon_t) \quad t \in \{0, 1, ...\}$$

$$k_0 \ given$$

where $\{\varepsilon_t\}_{t=0}^{\infty}$ is a sequence of random variables with conditional distribution function $f(\varepsilon_{t+1}|\varepsilon_t)$ for all t, and $\mathbb{E}_t x$ denotes the mathematical expectation of the random variable x conditional on information available at time t.

It is assumed that ε_t becomes known at t before c_t is chosen at t. The value of k_t is also known before c_t is chosen. This implies, through the transition equation, that k_{t+1} is known at t. Remark: In some specifications, the transition equation is of the form $k_{t+1} = G(k_t, c_t, \varepsilon_{t+1})$. In this case, ε_{t+1} and k_{t+1} are unknown at t. In other specifications, a stochastic shock is allowed to affect the return function: $u(k_t, c_t, \varepsilon_t)$.

The stochastic problem described above continues to have a recursive structure, which follows from the additive separability of the objective function in pairs (k_t, c_t), and from the particular form assumed for the transition equation. This implies that dynamic programming methods remain appropriate.

The Bellman Equation is:

$$V(k_t, \varepsilon_t) = \max_{c_t} u(k_t, c_t) + \beta \mathbb{E}_t \left\{ V(G(k_t, c_t, \varepsilon_t), \varepsilon_{t+1}) \right\}$$

where we have used the transition equation to eliminate k_{t+1}. Notice that the state vector is now (k_t, ε_t). The expectation is (assuming the shock is a continuous random variable):

$$\mathbb{E}_t\{V(G(k_t,c_t,\varepsilon_t),\varepsilon_{t+1})\} = \int V(G(k_t,c_t,\varepsilon_t),\varepsilon_{t+1}) f(\varepsilon_{t+1}|\varepsilon_t) d\varepsilon_{t+1}$$

By solving Bellman's functional equation we'll find the value function V, and *contingent plans* of the form:

$$c_t = h(k_t, \varepsilon_t)$$

and

$$k_{t+1} = g(k_t, \varepsilon_t) \equiv G(k_t, h(k_t, \varepsilon_t), \varepsilon_t)$$

The solution V of the Bellman Equation can be found by iterating on

$$V_{j+1}(k_t, \varepsilon_t) = \max_{c_t} u(k_t, c_t) + \beta \mathbb{E}_t \left\{ V_j(G(k_t, c_t, \varepsilon_t), \varepsilon_{t+1}) \right\}$$

starting from an appropriate function V_0 (*e.g.*, $V_0 = 0$).

For some cases it is also possible to use the Guess-Verify method.

We can also obtain the *stochastic Euler Equation*. The FOC give:

$$u_c(k_t, c_t)+\beta\mathbb{E}_t\left\{V_k(G(k_t, c_t, \varepsilon_t), \varepsilon_{t+1})G_c(k_t, c_t, \varepsilon_t)\right\} = 0$$

Then:

$$u_c(k_t, c_t)+\beta\mathbb{E}_t\left\{V_k(G(k_t, c_t, \varepsilon_t), \varepsilon_{t+1})\right\}G_c(k_t, c_t, \varepsilon_t) = 0$$

The equation above defines, implicitly, the contingency plan:

$$c_t = h(k_t, \varepsilon_t)$$

Substituting into the objective we get:

$$V(k_t,\varepsilon_t)=u(k_t,h(k_t,\varepsilon_t))+\beta\mathbb{E}_t\{V(G(k_t,h(k_t,\varepsilon_t),\varepsilon_t),\varepsilon_{t+1})\}$$

Differentiating with respect to k_t we get:

$$V_k(k_t,\varepsilon_t)=u_k(k_t,h(k_t,\varepsilon_t))+u_c(k_t,h(k_t,\varepsilon_t))h_k(k_t,\varepsilon_t)$$
$$+\beta\mathbb{E}_t\left\{\begin{array}{l} V_k(G(k_t,h(k_t,\varepsilon_t),\varepsilon_t),\varepsilon_{t+1})\times \\ \times[G_k(k_t,h(k_t,\varepsilon_t),\varepsilon_t)+G_c(k_t,h(k_t,\varepsilon_t),\varepsilon_t)h_k(k_t,\varepsilon_t)] \end{array}\right\}$$

$$=u_k(k_t,h(k_t,\varepsilon_t))$$
$$+\beta\mathbb{E}_t\{V_k(G(k_t,h(k_t,\varepsilon_t),\varepsilon_t),\varepsilon_{t+1})\}G_k(k_t,h(k_t,\varepsilon_t),\varepsilon_t)$$
$$+\left[\begin{array}{l} u_c(k_t,h(k_t,\varepsilon_t)) \\ +\beta\mathbb{E}_t\{V_k(G(k_t,h(k_t,\varepsilon_t),\varepsilon_t),\varepsilon_{t+1})\}G_c(k_t,h(k_t,\varepsilon_t) \end{array}\right]h_k(k_t,\varepsilon_t)$$

where the last term in the right-hand side of the second equality drops because of the FOC.

Then:

$$\begin{aligned} V_k(k_t,\varepsilon_t)= \quad & u_k(k_t,h(k_t,\varepsilon_t)) \\ & +\beta\mathbb{E}_t\{V_k(G(k_t,h(k_t,\varepsilon_t),\varepsilon_t),\varepsilon_{t+1})\}G_k(k_t,h(k_t,\varepsilon_t),\varepsilon_t) \end{aligned}$$

This is the Benveniste-Scheinkman (BS) formula (or Envelope Condition) for the stochastic case.

Suppose we manage to choose the state and control variables in such a way that the transition equation does not depend on k_t, so $k_{t+1} = G(c_t, \varepsilon_t)$. Then, $G_k = 0$ and the BS formula becomes:

$$\begin{aligned} V_k(k_t, \varepsilon_t) &= u_k(k_t, h(k_t, \varepsilon_t)) \\ &= u_k(k_t, c_t) \end{aligned}$$

Shifting the expression above one period ahead we get:

$$V_k(k_{t+1}, \varepsilon_{t+1}) = u_k(k_{t+1}, c_{t+1})$$

From the FOC:

$$u_c(k_t, c_t) + \beta \mathbb{E}_t \left\{ V_k(k_{t+1}, \varepsilon_{t+1}) \right\} G_c(k_t, c_t, \varepsilon_t) = 0$$

Combining the two expressions above we obtain the *stochastic Euler Equation*:

$$u_c(k_t, c_t) + \beta \mathbb{E}_t \left\{ u_k(k_{t+1}, c_{t+1}) \right\} G_c(k_t, c_t, \varepsilon_t) = 0$$

Appendix: The Euler Equation Again

The FOC gives:

$$u_c(k_t, c_t) + \beta V'(k_{t+1}) G_c(k_t, c_t) = 0$$

The BS formula gives:

$$V'(k_t) = u_k(k_t, c_t) + \beta V'(k_{t+1}) G_k(k_t, c_t)$$

From the expression above we get:

$$V'(k_{t+1}) = \frac{V'(k_t) - u_k(k_t, c_t)}{\beta G_k(k_t, c_t)}$$

Substituting into the FOC we get

$$u_c(k_t, c_t) + \beta \frac{V'(k_t) - u_k(k_t, c_t)}{\beta G_k(k_t, c_t)} G_c(k_t, c_t) = 0$$

which gives:

$$V'(k_t) = u_k(k_t, c_t) - u_c(k_t, c_t) \frac{G_k(k_t, c_t)}{G_c(k_t, c_t)}$$

Shifting the expression above one period ahead we get:

$$V'(k_{t+1}) = u_k(k_{t+1}, c_{t+1}) - u_c(k_{t+1}, c_{t+1}) \frac{G_k(k_{t+1}, c_{t+1})}{G_c(k_{t+1}, c_{t+1})}$$

Substituting the previous expression into the FOC we obtain the Euler Equation:

$$u_c(k_t,c_t) + \beta \left[u_k(k_{t+1},c_{t+1}) - u_c(k_{t+1},c_{t+1}) \frac{G_k(k_{t+1},c_{t+1})}{G_c(k_{t+1},c_{t+1})} \right] G_c(k_t,c_t) = 0$$

If we rewrite the transition equation $k_{t+1} = G(k_t, c_t)$ as $c_t = m(k_t, k_{t+1})$, the Euler Equation becomes:

$$\begin{aligned} 0 = {} & u_c(k_t, m(k_t, k_{t+1})) + \\ & \beta \left[\begin{array}{l} u_k(k_{t+1}, m(k_{t+1}, k_{t+2})) - \\ u_c(k_{t+1}, m(k_{t+1}, k_{t+2})) \frac{G_k(k_{t+1}, m(k_{t+1}, k_{t+2}))}{G_c(k_{t+1}, m(k_{t+1}, k_{t+2}))} \end{array} \right] G_c(k_t, m(k_t, k_{t+1})) \end{aligned}$$

which is a second-order difference equation in k. This equation can be solved using two boundary conditions: the initial condition k_0, and an appropriate transversality condition (TVC).

Bibliography

Sargent, Thomas. *Dynamic Macroeconomic Theory*, Harvard University Press, 1987. Chapter 1.

Stokey, N. and Lucas, R., with Prescott, E. *Recursive Methods in Economic Dynamics*, Harvard University Press, 1989.

A Simple Example

Choose c_0, c_1, x_1, and x_2, in order to solve:

$$\begin{array}{ll} \max & \ln c_0 + \ln c_1 + \ln x_2 \\ \text{s.t.} & x_1 = x_0 - c_0 \\ & x_2 = x_1 - c_1 \\ & x_0 > 0 \text{ given} \end{array}$$

Lagrange

$$\mathcal{L} = \ln c_0 + \ln c_1 + \ln x_2 + \lambda_1 (x_0 - c_0 - x_1) + \lambda_2 (x_1 - c_1 - x_2)$$

FOC

$$c_0 : \frac{1}{c_0} = \lambda_1$$

$$c_1 : \frac{1}{c_1} = \lambda_2$$

$$x_1 : \lambda_1 = \lambda_2$$

$$x_2 : \frac{1}{x_2} = \lambda_2$$

$$+ \, constraints$$

From the expressions above we get:

$$\frac{1}{c_0} = \frac{1}{c_1} = \frac{1}{x_2}$$

Then:

$$x_2 = c_1 = c_0$$

Substituting into the second constraint: $c_0 = x_1 - c_0 \Rightarrow$

$$x_1 = 2c_0$$

Substituting into the first constraint: $2c_0 = x_0 - c_0 \Rightarrow$ $3c_0 = x_0$

$$c_0^* = \frac{1}{3}x_0$$

Then: $x_2 = c_1 = c_0 \Rightarrow$

$$\begin{aligned} c_1^* &= \frac{1}{3}x_0 \\ x_2^* &= \frac{1}{3}x_0 \end{aligned}$$

Then: $x_1 = 2c_0 \Rightarrow$

$$x_1^* = \frac{2}{3}x_0$$

Dynamic Programming

At $t = 1$:

$$W_1(x_1) \equiv \max_{c_1, x_2} \ \ln c_1 + W_2(x_2)$$

$$\text{s.t.} \qquad x_2 = x_1 - c_1$$

$$x_1 \text{ given}$$

with $W_2(x_2) = \ln x_2$.

Then:

$$W_1(x_1) \equiv \max_{c_1, x_2} \ \ln c_1 + \ln x_2$$

$$\text{s.t.} \qquad x_2 = x_1 - c_1$$

$$x_1 \text{ given}$$

Then:

$$W_1(x_1) \equiv \max_{c_1} \ \ln c_1 + \ln(x_1 - c_1)$$

FOC:

$$\frac{1}{c_1} = \frac{1}{x_1 - c_1}$$

Then: $x_1 - c_1 = c_1 \Rightarrow$

$$c_1 = \frac{1}{2}x_1$$

Substituting into the constraint: $x_2 = x_1 - c_1 \Rightarrow x_2 = x_1 - \frac{1}{2}x_1 \Rightarrow$

$$x_2 = \frac{1}{2}x_1$$

Substituting into the objective: $W_1(x_1) = \ln(\frac{1}{2}x_1) + \ln(\frac{1}{2}x_1) \Rightarrow W_1(x_1) = 2\ln(\frac{1}{2}x_1) \Rightarrow$

$$W_1(x_1) = 2\ln(\frac{1}{2}) + 2\ln x_1$$

At $t = 0$

$$W_0(x_0) \equiv \max_{c_0, x_1} \ \ln c_0 + W_1(x_1)$$

$$\text{s.t} \qquad x_1 = x_0 - c_0$$

$$x_0 \text{ given}$$

where $W_1(x_1)$ was found in the previous step.

Then:

$$\begin{aligned} W_0(x_0) \equiv \max_{c_0, x_1} \ & \ln c_0 + 2\ln(\tfrac{1}{2}) + 2\ln x_1 \\ \text{s.t.} \qquad & x_1 = x_0 - c_0 \\ & x_0 \text{ given} \end{aligned}$$

Then:

$$W_0(x_0) \equiv \max_{c_0} \ \ln c_0 + 2\ln(\frac{1}{2}) + 2\ln(x_0 - c_0)$$

FOC:

$$\frac{1}{c_0} = \frac{2}{x_0 - c_0}$$

Then: $x_0 - c_0 = 2c_0 \Rightarrow 3c_0 = x_0 \Rightarrow$

$$c_0^* = \frac{1}{3}x_0$$

Substituting into the constraint: $x_1 = x_0 - c_0 \Rightarrow x_1 = x_0 - \frac{1}{3}x_0 \Rightarrow$

$$x_1^* = \frac{2}{3}x_0$$

Substituting the expression above into $c_1 = \frac{1}{2}x_1$ we get: $c_1 = \frac{1}{2}\frac{2}{3}x_0 \Rightarrow$

$$c_1^* = \frac{1}{3}x_0$$

Substituting $x_1 = \frac{2}{3}x_0$ into $x_2 = \frac{1}{2}x_1$ we get:

$$x_2^* = \frac{1}{3}x_0$$

Notice that both methods give the same solution.